

Examiner reports

Q1.

Here most correctly identified the end points of the interval. Many included the end points in their interval and so lost the accuracy mark.

Q2.

This lengthy question tested a range of skills and gave students the opportunity to show how well they had mastered these. There were few who could not pick up some marks here.

For part (a), most started off with a correct combination of the $x\sqrt{x}$ into a single power and differentiated the expression correctly. It was expected that at this stage they would explain that for a stationary value the gradient was 0. Many simply used this fact.

Here one approach was to solve the equation and select $x = 1$. This approach did the groundwork for part (b). Alternatively, they could verify that $x = 1$ resulted in a gradient of 0. This approach was easier here but meant that more work was needed in part (b).

Use of the second differential was needed to prove that this was indeed a maximum. Finally, it was necessary to show that $x = 1$ gave $y = 3$. Many omitted this last step and so also lost the R1 mark, which required clear mathematical rigour throughout. However, simply failing to explain that for a stationary value the gradient was 0 did not result in the loss of the R mark.

For part (b), as an alternative to algebraic methods, it was permitted for students to use graphical calculators to identify this second stationary point.

Q3.

This question gave students complete choice as to how to go about solving the problem. Many good attempts were seen, with the full range of marks being scored. Different correct approaches were used and given credit. For example, some students modelled using the length and width of the rectangle, others used a radius and angle to form an expression for area using trigonometry.

Marks were often lost through students not fully justifying their answer. Some students failed to identify their variables or were inconsistent in their use. Others did not explain why they had differentiated or how they knew they had found a maximum.

Q4.

Differentiation from first principles is new to the A-level maths specification and while many students seemed well practiced in the routine of part (a), only the strongest students could correctly explain how the gradient of the chord led to A being a stationary point.

The most frequent mistake seen in part (a) was the incorrect manipulation of negatives in the expansion of $(-4 + h)^3$. Some attempts were too piecemeal and did not form coherent arguments.

Around half of the students completed part (b) successfully.

The most common mistake in part (b) was to simply $h = 0$, rather than explain that as h tends to zero the gradient of the chord approached the gradient of the tangent. The most successful students simply used correct limit notation to show that the gradient of the curve was zero at A and hence a stationary point.

Q5.

Again, the vast majority of students answered this multiple choice question correctly, giving the answer 7.

Q6.

In (a), most students realised that the domain would take values $x > 1$ in this question. However, in order to gain full marks it was necessary to state the domain in a complete form involving the notation $x \in \mathbb{R}$, which most students failed to do.

In (b), a large number of students started to differentiate the printed result for $f'(x)$, rather than differentiate $f(x)$. Many then realised their mistake and restarted the question on a continuation sheet, with a good number scoring full marks.

There was a wide spread of marks across question (c). Most students understood that they needed $f''(x) = 0$, though a number instead used $f'(x) = 0$. Of those students who successfully solved the correct equation, very few scored full marks, as they either failed to correctly deal with the solution $x = 1$ or failed to test either side of their correct solution $x = 4$. Only a small proportion of students identified the correct interval in (d).

Q17.

(a) The differentiation was almost always correct.

(b) (i) Having only a single derivative to select meant that the success rate in finding the rate of change was greater than some years. Nevertheless, some had difficulty simplifying $\frac{1}{2} - 2$ without a calculator.

(ii) Most students made the correct deduction from their value in part (b)(i) but often the explanations were not clear enough to earn the mark. It was

necessary to write something like “ $\frac{dy}{dt} < 0$, therefore the “height is decreasing”, making specific reference to the **negative** value of the derivative.

(c) (i) Once again, the second derivative was usually found correctly, but arithmetic errors sometimes arose when finding the value of $\frac{d^2y}{dt^2}$ when $t = 2$.

(ii) The interpretation of the sign of $\frac{d^2y}{dt^2}$, as indicating whether y had a minimum or maximum value, seemed to be well known.

Q18.

The different format of this question took many students by surprise. Many weaker students were confused about the processes required. Some even used integration in part (a) and gradients in part (b) or thought that they were required to find the normal and the tangent in the two parts.

- (a) The majority realised the need to substitute $x = 1$ into the given expression for $\frac{dy}{dx}$, but even that was not always evaluated correctly. Quite a few differentiated the given expression and scored no marks at all. Although most went on to find the equation of the tangent in the required form, a substantial number found the normal instead. Others hedged their bets by finding the equations of both the tangent and the normal, with no indication as to which was which. It was disappointing to see that several students, who obtained a correct equation for the tangent in the form $y - 4 = 9(x - 1)$, were unable to simplify this correctly to the form $y = 9x - 5$.
- (b) When integration was attempted, it was usually done correctly with just the odd mistake in signs or coefficients. However, many failed to include the constant of integration, “+ c”, and could proceed no further. Many who did have the “+ c” stopped at that point. Some substituted $x = 1$ into their expression for x and equated this to zero. This could not be given any credit because they did not have an expression of the form $y = \dots$ and so were unable to substitute $y = 4$. Many of those who found the correct value of the arbitrary constant failed to write the final equation in the form $y = f(x)$.

Q19.

In general, this question was a good source of marks for many students. Most students were able to find the correct expression for $\frac{dy}{dx}$ in part (a). In part (b), the majority of students were able to obtain the printed answer in a convincing manner by firstly clearly indicating that they were finding the value of $\frac{dy}{dx}$ at $x = 2$. There were, however, some cases of students who obtained the printed answer by clearly incorrect methods and some by showing incomplete details; both of which were penalised heavily. In part (c)(i), those students who correctly approached the problem by trying to solve the equation $\frac{dy}{dx} = 0$ generally picked up 2 of the 3 marks for finding the correct coordinates of M but quite frequently then forgot to state the conclusion, that the stationary point lay on the x -axis.

In a number of solutions, students, having solved $\frac{dy}{dx} = 0$ correctly to get $x = -2$, just stated that the point M lay on the x -axis. In part (c)(ii), those students who produced a rough sketch were usually more successful in recognising that the x -axis was the required tangent than those who used the general equation of a line. In this context, it was not always the case that students realised that the product of zero and $(x + 2)$ was 0. Those students who attempted part (d) generally understood the method of solution but the required coordinates of T were frequently incorrect.

Q20.

- (a) (i) Almost all candidates attempted to find $p(-3)$ and most were successful in showing that $p(-3) = 0$. It was pleasing to see many candidates taking heed of previous reports and writing a concluding statement such as “so $x + 3$ is a factor”. Those who simply used long division were ignoring the request to use the Factor Theorem and earned no marks.
- (ii) The majority of candidates used inspection or long division and found the correct quadratic factor. A few using the latter method did not always write their final answer as the product of factors as requested.
- (b) (i) Almost all found the first derivative correctly, though a few added “+ c ”.
- (ii) Several candidates presented an inadequate solution here. It was necessary to both equate their answer to part (b)(i) to zero and divided by 4 in order to obtain the given equation.
- (iii) Solutions to this part were very disappointing. Most candidates merely substituted $x = -3$ into the given equation and were, in effect, repeating part (a). Those who attempted to solve the quadratic equation or found the value of the discriminant did not always give a satisfactory explanation for the single stationary point of the curve. For example “the discriminant is negative therefore the only real root is $x = -3$ ”.
- (iv) Once again the second derivative was found correctly by most, although its value was sometimes incorrect as a result of poor arithmetic, particularly in the evaluation of the term $12(-3)^2$. Unfortunately, quite a few candidates divided their expression for $\frac{dy}{dx}$ by 4 before differentiating again and wrote $\frac{d^2y}{dx^2} = 3x^2 - 4$, which was incorrect.
- (v) The interpretation of the sign of the second derivative was well known. However not all candidates gave an adequate reason. A statement such as “ $92 > 0$ ” or “ $\frac{d^2y}{dx^2}$ is positive” was required; an answer such as “it is positive” or “the value is greater than 0”, without specific reference to the value from part (iv), was not sufficient.

Q21.

A majority of students could cope with the powers in part (a) but a significant proportion of others ended up with the wrong second term by thinking that the denominator x should be placed under both the x^2 and the $x^{1/2}$ terms resulting in a common error of the second term being given as $x^{1/2}$ a rather than $x^{3/2}$. Most students showed that they could differentiate accurately and a large majority showed that they knew the method for finding the equation of the normal to a curve, with many scoring at least 3 of the 4 marks in part (b)(ii). The final part of the question, which introduced the stationary point, usually gave average students the opportunity to pick up the first mark but in general only the most able students could then cope with the fractional indices to correctly reach $x^{5/2} = 8$ and then to solve this to show that the x -coordinate of the stationary point could be written as $2^{6/5}$.

Q22.

This question was well answered by the great majority of the students.

- (a) Any errors that occurred were usually associated with the second term where $\tan 2x$ was differentiated to $\sec^2 2x$ or $2\sec^2 x$ instead of $2\sec^2 2x$. Attempts at using the quotient rule on $y = \frac{x^4 \sin 2x}{\cos 2x}$ was often prone to errors.
- (b) Most students used the quotient rule correctly, although some had the two terms in the numerator reversed. Those who chose to use the product rule were mostly correct, with a few making a sign error. A small number of students failed to go on substituting $x = 3$ into their derived equation. A common incorrect answer was $\frac{3}{2}$; arising from students expanding the denominator incorrectly to $x^2 - 2x - 1$.

Q23.

This question was generally done well with over 50% of the entry scoring 9 or more of the 12 marks available. Most students lost marks in part (c) by attempting to use an inefficient method.

- (a) Virtually all students used parametric differentiation and most had the two required derivatives correct. Occasionally there was a coefficient error or a t was brought down, but most students knew the chain rule and used it appropriately. Most students had the correct unsimplified result, which was accepted for the marks, but some made an error in simplifying, usually dropping the negative sign and so lost two marks in part(b). Some students 'simplified' $1 / 16e^{-2t}$ to $16e^{2t}$.
- (b) (i) Most students had this correct, either by use of their calculator or by hand, although there was evidence that some students were not using the exponential button on their calculator correctly.
- (ii) Although again most students had these coordinates correct, a number dropped the 4 from one of the given equations, usually from the x equation, so lost a mark here and another in part (iii) as they could not then get the correct answer.
- (iii) Most students used their results from parts (i) and (ii) to set up a correct equation of the normal, either using $y - y_1 = m(x - x_1)$ or $y = mx + c$. Some went on to complete the question successfully but others confused the axes and substituted $x = 0$ and found y . Some students misunderstood the question and substituted $y = 0$ into the parametric equation for y and found t , seemingly ignoring the reference to the normal given in the question.
- (c) There were many different approaches to finding the Cartesian equation. The most straight forward way was to substitute into the given expression and show that it simplified to the requested constant. The common error in this approach was a sign error, often resulting in $k = 0$. The result $e^{2t} e^{-2t} = 1$ was generally well known although occasionally the product became $e^{\pm 4t}$. The other fairly common approach was to solve one of the equations for $e^{\pm 2t}$ and substitute in the other equation, and then simplify. This often resulted in algebraic errors. Some students carried out an unnecessary further stage and actually solved for t in terms of x and / or y using logarithms. This often resulted in some rather cumbersome expressions when substituting back and students made many sign and algebraic errors in trying to

convert their logarithm expressions into the requested form.

A typical error was $e^{-\ln\left(\frac{x+4}{8}\right)} = -\left(\frac{x+4}{8}\right)$. Some students did however derive the required result through insightful manipulation of the parametric equations.

Q24.

In part (a), differentiation had been well drilled and most candidates earned all 4 marks for the two derivatives. Occasionally, the + 1 was omitted or an extra 5 or + x was included, and a few felt the need to add + c to both their answers.

In part (b), those candidates who substituted $x = -1$ into their expression for $\frac{dy}{dx}$ often obtained a gradient of 12. However, not all used this value: some took the negative reciprocal and then formed the equation of the normal. Some candidates did not use their expression for $\frac{dy}{dx}$ in order to find the gradient but erroneously used the gradient of AB , with value 2, instead.

Surprisingly, a very small minority earned full marks in part (c). Most candidates verified the nature of the stationary point by considering the sign of the second derivative. However only a handful recognised the need to verify that the point was stationary in the first place, by showing that the first derivative was equal to zero.

In part (d)(i), most candidates were able to integrate the expression, with only the weakest candidates not making any attempt. Poor notation was used, with many including the integral sign after integrating. However dealing with the substitution of -1 caused many errors. Manipulation of fractions caused problems too; it was disappointing to see many candidates with correct fractions being unable to combine these to give a value of 8 or equivalent. Many candidates did not find the actual value of the definite integral until part (d)(ii), and on this occasion full credit was given, but this may not be the case in future.

In part (d)(ii), most candidates calculated the area of the relevant triangle to be 2 and subtracted this from their answer to part (d)(i). Errors included writing the area of the

triangle $\frac{4 \times 2}{2}$ as and careless arithmetic such as $8 - 2 = 4$.

Q25.

This final question proved to be less demanding than has been the case in many of the past papers for this unit. Even the unstructured final part of the question produced a significant number of fully correct solutions from the candidates. Parts (a) and (b)(i) were found to be very straightforward with most candidates scoring heavily. It is worth recording that a number of candidates in part (b)(ii) failed to give sufficient justification for taking the gradient m to be -8 . With the answer given, it was not acceptable to just state that the

gradient is -8 without some clear indication that this value came from the value of $\frac{dy}{dx}$ when $x = 8$. In part (c), the most common loss of a mark was through failing to simplify

$\frac{3}{\left(\frac{8}{3}\right)}$ correctly. The error of integrating $x^{\frac{5}{3}}$ to get a term in the form $kx^{\frac{7}{3}}$ was penalised

more heavily.

In part (d), those who found the area of the triangle by solving the equations of OP and AP simultaneously to find the coordinates of P and then using $A = \frac{1}{2}bh$ were generally much more successful than those who attempted to use integration, as their limits were frequently incorrect: 0 to 8 in both cases rather than 0 to 3.2 for OP and then 3.2 to 8 for AP . It was also surprising to see a significant minority of candidates finding the lengths of OP and AP and the angle OPA and then using $\frac{1}{2}ab \sin C$ to find the area of the triangle; although a valid approach it took much more time and usually led to rounding errors. The method for finding the area under the curve was well understood but errors in part (c) frequently resulted in the area under the curve being greater than the area of the triangle: a clear error which should have been spotted by reference to the diagram.

Q26.

Most of the candidates had part (a) correct, with the majority choosing to substitute $t = \frac{3}{y}$ into the given equation for x rather than to substitute into the given cartesian equation. The commonest, but rarely seen, error was to not square the 3 or to make a sign error.

The majority of candidates chose to approach part (b)(i) through parametric differentiation, and most did it accurately, with many gaining full marks. The most common error was to lose the minus sign on the -16 gradient, having initially found this correctly. Others made mistakes with the fractions.

For part (b)(ii), there are numerous ways in which this intersection can be verified, and many different ways were attempted by the candidates. Many scored 1 mark by finding the y - value, either directly from their tangent or from the curve, or even via the parametric equations.

Some substituted their tangent equation into the curve and attempted to verify that $x = \frac{3}{2}$ satisfied the equation. Some did this convincingly, but others proceeded rather more in hope than expectation. Some tried to factorise the resulting cubic equation. The most efficient method was to find the y -value from the equation of the tangent and then verify that $(\frac{3}{2}, -8)$ satisfies the equation of the curve. This method was mostly seen from those who achieved full marks on the question.

Q27.

In part (a)(i), some students merely doubled the given equation and then divided by 2; this did not convince examiners and earned no marks. Students were expected to show the separate areas of the faces of the cuboid; the minimum working required was sight of $2xy + 6xy$ as well as $6x^2$, before combining these to give $6x^2 + 8xy = 32$ and hence the printed result. Some proved the result by considering half the surface area and, provided enough explanation was given, this also scored full marks.

In part (a)(ii), the majority earned a single mark for writing down a correct expression for the volume in terms of x and y . The algebra required to rearrange the equation from part (a)(i) to make y or xy the subject proved too difficult for many and once again highlighted students' algebraic weaknesses as they tried to reach the printed answer but violated various algebraic laws on the way.

In part (b), whilst the term $12x$ was almost invariably differentiated correctly, the same could not be said of the second term in the expression for V , and $-27x^2$ was a common incorrect answer.

In part (c)(i), although most students realised the need to substitute $x = \frac{4}{3}$ into their expression for $\frac{dy}{dx}$, not all showed the required working to prove that $\frac{dy}{dx} = 0$. It was not

sufficient to write $12 - \frac{27\left(\frac{4}{3}\right)^2}{4} = 12 - 12 = 0$; evidence of correct manipulation of the fractions was needed. Once again, many students did not write a concluding statement with regard to the zero gradient implying that there was a stationary point.

In part (c)(ii), by far the most common error when finding $\frac{d^2y}{dx^2}$ was the omission of the minus sign. The final mark was still available even if students made errors in finding the second derivative, provided the substitution of $x = \frac{4}{3}$ was done correctly and the appropriate conclusion was drawn regarding the nature of the stationary point.

Q28.

In part (a)(i), the idea of y being increasing when $\frac{dy}{dx} > 0$ seemed unfamiliar to many students, who seemed content to substitute a few numerical values for x into the expression for $\frac{dy}{dx}$ rather than tackling the three-line proof. Consequently, very few scored full marks on this part.

Many students ignored the inequality signs throughout part (a)(ii). It was pleasing to see many students attempting to factorise the quadratic, often successfully, rather than relying on the formula. Perhaps more students would be successful in solving quadratic inequalities if they were to draw a sign diagram or sketch a graph. There were many correct solutions, but students must realise that $x < 2$, $x > \frac{4}{3}$ is not the same as $\frac{4}{3} < x < 2$.

In part (b)(i), most students attempted to evaluate $\frac{dy}{dx}$ when $x = 2$. Not all were able to cope with the evaluation of $-6(2)^2$, which they needed to do in order to convince examiners that $\frac{dy}{dx}$ was indeed equal to zero. Once more, some failed to write a concluding statement or referred to a stationary point rather than the fact that the tangent

was parallel to the x -axis.

In part (b)(ii), it was pleasing to see some fully correct solutions. Most recognised that the equation of the tangent at P was $y = 3$, but it was disappointing to see a number of students write $y - 3 = 9(x - 2)$ followed by $y = x + 1$ as their tangent equation. In finding

the equation of the normal at Q , some, having correctly found that $\frac{dy}{dx} = 10$ when $x = 3$,

used the value 10 instead of $-\frac{1}{10}$ for the gradient of the normal. Arithmetic errors were very common by those who used $y = mx + c$ when finding the value of c ; typically $y = \frac{x}{10} + \frac{7}{10}$ was seen instead of

$y = \frac{x}{10} - \frac{13}{10}$. Curiously, quite a few students who had previously obtained $y = 3$ as the equation of the tangent, and consequently the correct y -coordinate of the point R , then substituted the correct value of x into their incorrect equation for the normal and found a different, and now incorrect value for the y -coordinate of R .

Q29.

In part (a)(i) of this calculus question, where students were asked to verify that a given expression was equal to zero when $x = 2$, a minority of students failed to give adequate written evidence, particular when the order of calculation was important; the minimum

examiners accepted here was $3 \times 4 - \frac{4}{4} - 11 = 0$, although $12 - 1 - 11 = 0$ was more

convincing. In part (a)(ii) the differentiation of $-\frac{4}{x^2}$ was a hurdle not always overcome by the weaker students. However, those who first re-wrote it as $-4x^{-2}$ usually went on to differentiate it correctly. Many students scored the mark for giving a correct reason in part (a)(iii). In part (b) a large number of the students did realise that they needed to integrate the given expression for the gradient to find the equation of the curve but a significant number of them did not include the arbitrary constant which is essential here, and only a minority knew how to find its value from the given information. Although only a minority, it was still disappointing to see more students than expected writing down the equation of the curve in the form $y - y_1 = m(x - x_1)$.

Q30.

This question was in general a good source of marks for the majority of students. The main cause of loss of marks was a common desire to use an approximate numerical value for e .

Q31.

Part (a) proved to be a good source of marks, although a significant number of students

lost a mark through poor notation, such as $\cos y^2$ or no $\frac{dx}{dy}$

Although many earned full marks in part (b), many other proofs were inadequate.

Students were expected to use the identity $\tan^2 y + 1 = \sec^2 y$ and to show the expansion

of $(x - 1)^2$ clearly. Too many fudged their working, wrote $\tan y + 1 = \sec y$ or only proved the known trig identity.

The result in part (c) was easily derived from the previous two parts, but there were many blanks here and many confused statements about inverse functions which were often

taken as $\frac{1}{\tan y}$. Students who used the formula book appropriately generally earned the mark.

There were many fully correct solutions to part (d)(i). However it was distressing at this stage to see some horrendous algebra, with terms inverted in ways that were invalid and worthless.

Only the more able students were able to make progress on the last two parts of the question. Many students differentiated their quadratic instead of the expression for $\frac{dy}{dx}$. Some students earned the method mark in the final part, but few completed the whole answer, as their final explanation often had an element missing.

Q32.

For part (a)(i), most students knew that they needed to differentiate the two parametric equations and use the chain rule which many did correctly, with very few having it the wrong way up. Errors were relatively rare but included sign errors, coefficient errors or dropping the 2 from the double angle. Some chose to expand $\sin 2\theta$ and use the product rule, again mostly successfully, but some made sign errors here too.

The manipulation to the required form was often confused as some students lost their way, many of these immediately replacing the $\frac{1}{\sin \theta}$ term from differentiation by $\operatorname{cosec} \theta$, and abandoning, or fudging, as they could not get to the required form. Those who immediately replaced $\cos 2\alpha$ with $1 - 2 \sin^2 \alpha$ usually went on to be successful. The common error was to omit the 2 in the expansion. Those who went via the $\cos^2 \theta - \sin^2 \theta$ form were sometimes successful, although many made a sign or coefficient error in attempting to simplify to the required form.

Part (a)(ii) was essentially testing a student's ability to use $\theta = \frac{\pi}{6}$ correctly in their version of $a \sin \alpha + b \operatorname{cosec} \alpha$ in finding the gradient of a tangent and the associated normal. Many did it correctly, although a significant number only found the gradient of the tangent whereas some unnecessarily found the equation of the normal. Some students used their chain rule expression, which was perfectly acceptable. The relationship between the gradients was generally very well known.

Many students made little or no progress with part (b), especially if they tried to start from inverse sines and cosines from the parametric equations. Those who expanded $\sin 2\theta$ and then squared the whole equation usually progressed to trying to use the parametric equations to eliminate θ , but relatively few could then manipulate correctly to the required form. Few scored high marks here, but some very insightful and clever derivations were seen, based on $\sin^2 \theta + \cos^2 \theta = 1$.

Q33.

Although this question had no structure to it, most students realised it involved implicit differentiation and attempted to find an expression for $\frac{dy}{dx}$, with most not realising this was in fact unnecessary, as to make progress in solving the problem they only needed to put their $\frac{dy}{dx}$ terms equal to zero. Many students got the implicit differentiation fully correct, some using a spurious $\frac{dy}{dx}$ on the left hand side, but only using it as an aide memoire, which was condoned on this occasion, whereas a few brought it into their expression and this was penalised. Errors were seen in all the terms but commonly the 3 did not go to zero, $\frac{dy}{dx}$ was missing from the y -term or there was a coefficient error. Sign errors were also made in using the product rule, which appeared to be related to a failure to use brackets, and some students gave this derivative as a single term..

Many did solve their derivatives expression correctly for $\frac{dy}{dx}$, but when equated to zero some did not continue any further, some equated both numerator and denominator to zero, and some multiplied across, treating the zero as if it were 1. Some just guessed values for x and y at this stage, making the numerator come to zero. Credit was only given for equating the numerator to zero, and obtaining y in terms of x , or vice versa, so that a solution could be continued if students realised they now needed to substitute back into the original equation. Some did this correctly, and some fully correct solutions were seen, but others pursued equations obtained through incorrect algebra, and gained no further credit. Some students who had progressed correctly found only the positive value of x , so could not find two stationary points. Some others who had correctly found two values of x , substituted into the original equation, rather than the equation from the derivatives, and found a spurious value of y .

Q34.

In part (a) it was good to see almost everyone being able to differentiate correctly.

A few then went on to divide their expression for $\frac{dy}{dx}$ by 2 or 3 or 6, but this was not penalised at this stage.

In part (b) most candidates tackled the verification that P was a stationary point first and generally did so correctly, although some did not present accurate working, writing things such as $18 + 6 - 12 = 0$. Many stopped at this point, although a few tried listing

various values of x with the corresponding value of $\frac{dy}{dx}$, and occasionally discovered

the other value of x for which y was stationary. Many candidates equated $\frac{dy}{dx}$ to zero, but not all were then able to factorise the quadratic correctly. Some felt the need to use the quadratic equation formula and were not always successful in finding the correct solutions.

Part (c)(i) Those who had the correct expression for $\frac{dy}{dx}$ usually found the second

derivative correctly. Many ignored the request to find the value of $\frac{d^2y}{dx^2}$ when $x = -1$, but this was sometimes generously credited in part (c)(ii), where it was pleasing to see most candidates being able to interpret the sign of the second derivative in order to describe the nature of the stationary point.

Q35.

In part (a)(i) candidates who recognised that the first step required was differentiation generally produced correct answers, but a few had trouble with the “ $-x$ ” term and others were unable to combine $-1 + (-4)$ correctly. Some weaker candidates assumed that they could find the gradient using the coordinates of two points.

In part (a)(ii) once again there was much confusion between parallel and perpendicular lines, with many candidates giving the wrong gradient of the tan gent. To earn the final mark it was essential to express the equation in the specified form and some failed to do so; for instance, an answer such as $y + 5x = 17$ is not in the correct form.

In part (b)(i) the integration was generally correct and most candidates made an attempt to substitute the correct limits. A few however could not integrate “14” and left it as a constant. Very few were able to combine their fractions correctly and consequently only a small proportion of the candidates found the correct value of the integral.

In part (b)(ii) candidates who had only found the definite integral in part (b)(i) were generously rewarded with the remainder of the marks here. Not everyone recognised the need to find the area of a triangle in order to find the area of the shaded region. A few correctly found the area under the line $y = 4x + 8$ by integration, but this is not the recommended approach.